THE EXT GROUPS OUT OF A LINEARLY ADEQUATE FUNCTOR VANISH IN POSITIVE DEGREES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The presentation of a linearly adequate functor by finite free module-valued functors. The definition of linearly adequate functors used to produce P. Supplies the exact sequence 0 -> L -> A^m -> A^n -> P -> 0.
- Vanishing of positive Ext groups out of a finitely presented module-valued functor. Same chapter, Lemma lemma-ext-group-zero-key, invoked at proof lines 1573-1579. Gives the base cases for P and for the free module.

Source reference. Original lemma 06ZG: sources/stacks-adequate/pinned-original-source.tex, source lines 1425–1425. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring. Let M be an A-module. Let L be a linearly adequate functor on Alg_A. Then $\mathrm{Ext}^i_{\mathcal{P}}(L, \underline{M}) = 0$ for $i > 0$ where $\mathcal{P}$ is the category of module-valued functors on Alg_A.*

Proof. Since L is linearly adequate there exists an exact sequence

$$0 \to L \to \underline{A^{\oplus m}} \to \underline{A^{\oplus n}} \to \underline{P} \to 0$$

Here $P = \mathrm{Coker}(A^{\oplus m} \to A^{\oplus n})$ is the cokernel of the map of finite free A-modules which is given by the definition of linearly adequate functors. By Lemma 06ZG we have the vanishing of $\mathrm{Ext}^i_{\mathcal{P}}(\underline{P}, \underline{M})$ and $\mathrm{Ext}^i_{\mathcal{P}}(\underline{A}, \underline{M})$ for $i > 0$. Let $K = \mathrm{Ker}(\underline{A^{\oplus n}} \to \underline{P})$. By the long exact sequence of Ext groups associated to the exact sequence $0 \to K \to \underline{A^{\oplus n}} \to \underline{P} \to 0$ we conclude that $\mathrm{Ext}^i_{\mathcal{P}}(K, \underline{M}) = 0$ for $i > 0$. Repeating with the sequence $0 \to L \to \underline{A^{\oplus m}} \to K \to 0$ we win. □

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